

Student Performance Analysis

Education | Academic Year 2024

Total Students	Pass Rate	Avg Score	Avg Attendance	Avg Study Hrs	At-Risk
1500	99.9%	76.4 / 100	78.2%	4.0 hrs / day	129

Executive Summary

This report examines academic performance data for 1,500 students across five subjects. The analysis measures pass rates, identifies subject-level strengths and weaknesses, quantifies the impact of attendance and study hours on exam scores, and flags students at risk of failing. The objective is to provide educators with actionable data to improve outcomes before the end of the academic year.

Key Insights

- Overall pass rate is 99.9%, with 129 students flagged at risk.
- Science is the strongest subject with an average score of 76.7.
- History is the weakest subject with an average score of 76.1.
- Attendance and score have a positive correlation of $r = 0.338$.
- Study hours and score show a moderate correlation of $r = 0.270$.
- Section C has the lowest pass rate across all sections.

Recommendations

- Launch targeted remedial classes for History immediately.
- Enforce a minimum 75% attendance policy and track violations weekly.
- Assign peer tutors in Section C to support weaker students.
- Encourage all students to commit to at least three study hours per day.
- Conduct weekly check-ins for the 129 identified at-risk students.
- Share best practices from top-performing sections with all teachers.

Figure 1

Overall Pass / Fail Distribution

The chart shows the proportion of pass and fail outcomes across all five subjects and 1,500 students. A pass rate of 99.9% indicates most students are meeting the 40-mark threshold. Tracking this figure each term provides a quick headline measure of institutional performance.

Pass Rate	Fail Records	Total Records
99.9%	5	7,500

Overall Pass / Fail Distribution

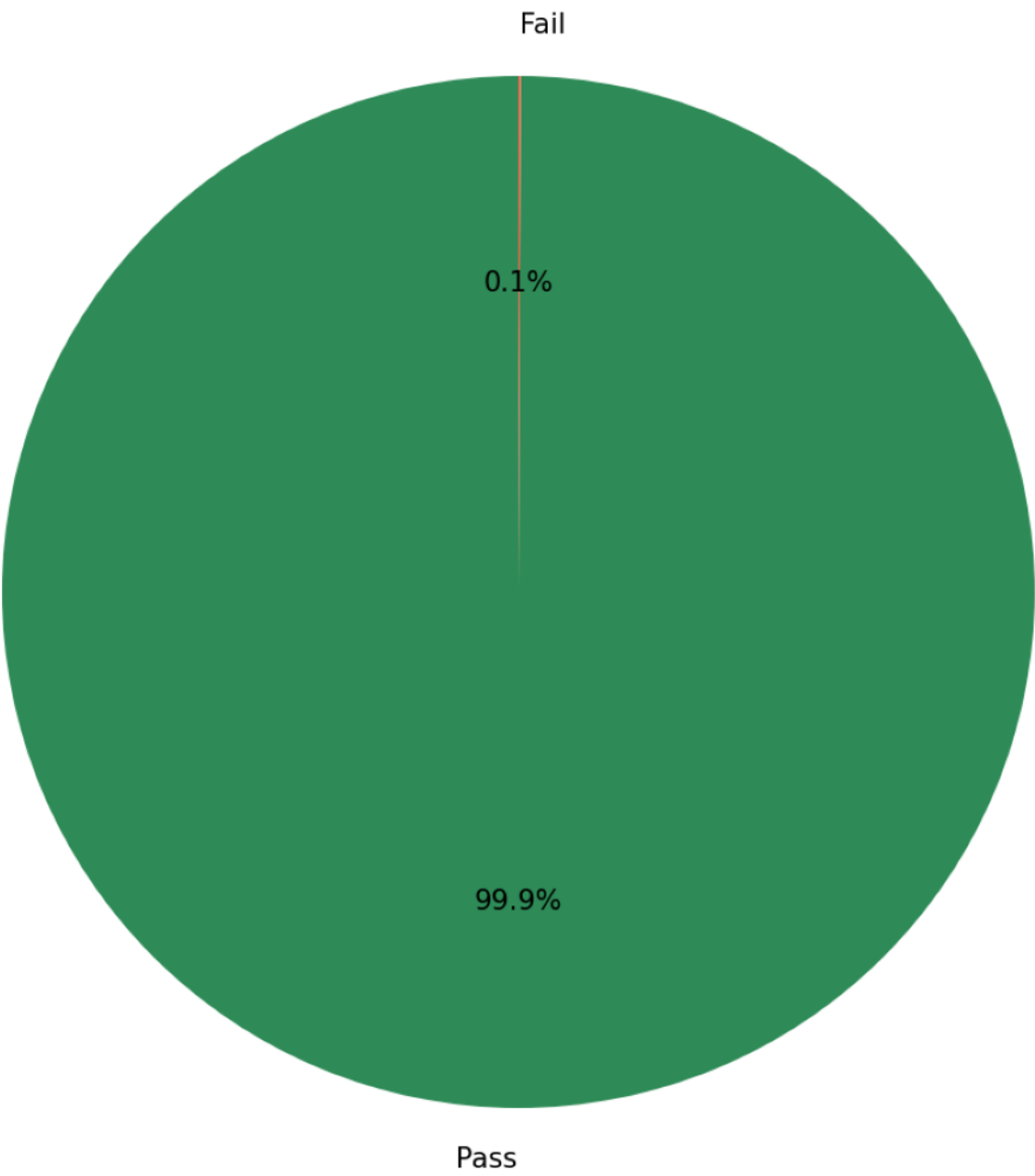


Figure 2

Average Score by Subject

Scores vary across subjects. Science leads at 76.7 marks on average while History trails at 76.1. These gaps highlight where curriculum delivery or student engagement needs attention. Subjects with consistently lower averages may benefit from revised teaching approaches or additional practice material.

Science	Maths	English	CS	History
76.7 / 100	76.5 / 100	76.4 / 100	76.3 / 100	76.1 / 100

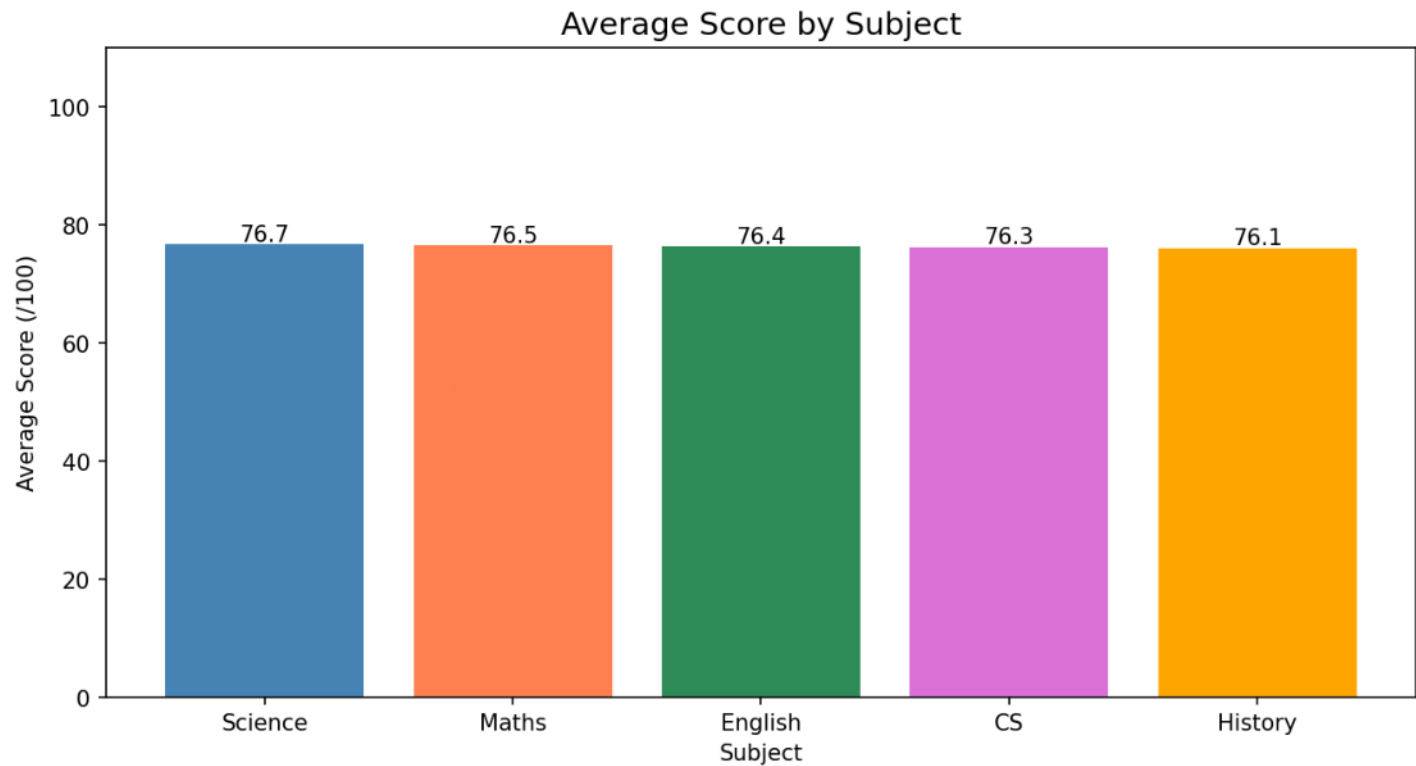


Figure 3

Score Distribution by Subject

KDE curves show the shape of score distributions per subject. A narrow tall peak means scores cluster tightly; a wide flat curve indicates spread. The dashed line at 40 marks is the pass threshold. Subjects where a visible portion of the curve falls below this line carry a higher fail risk.

Pass Threshold	Subjects	Method
40 / 100	5	KDE

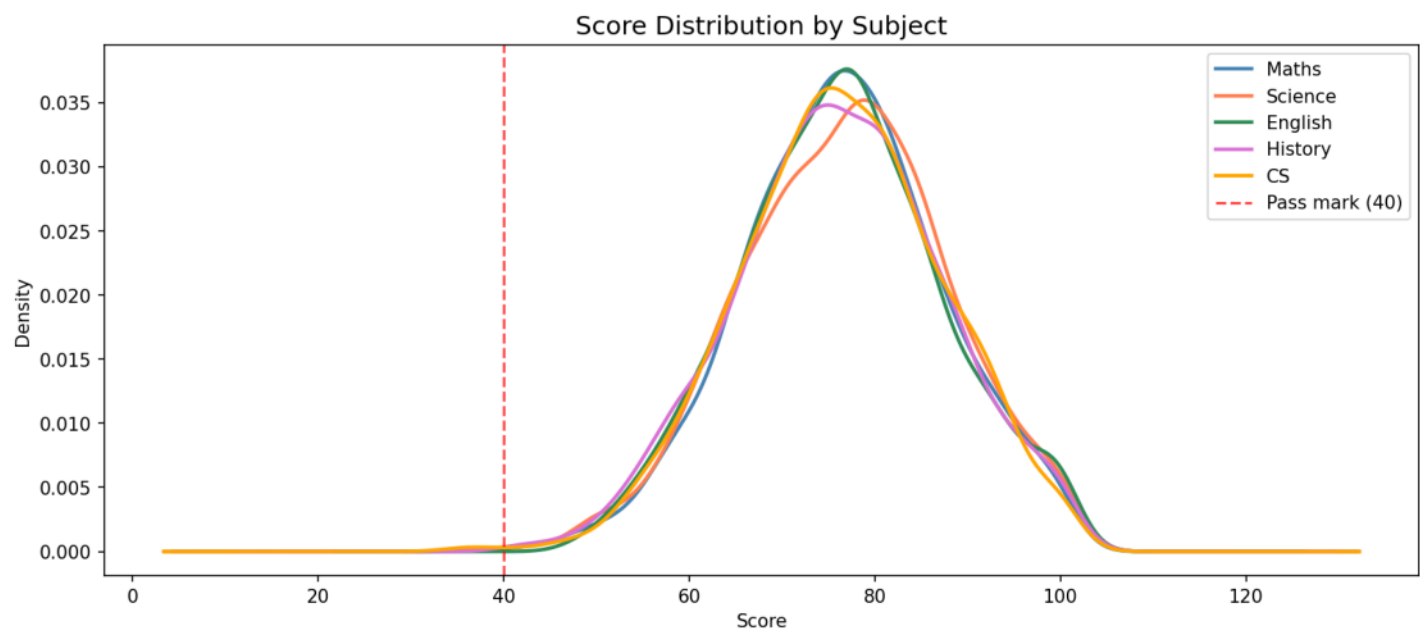


Figure 4

Attendance vs Score

Each point is one student-subject record — green for pass, red for fail. The trend line confirms a positive relationship ($r = 0.338$): students with higher attendance tend to score higher. Most fail cases cluster in the lower-left where both attendance and scores are low.

Attendance-Score r	Avg Attendance	Below 60% Attend.
0.338	78.2%	129 students

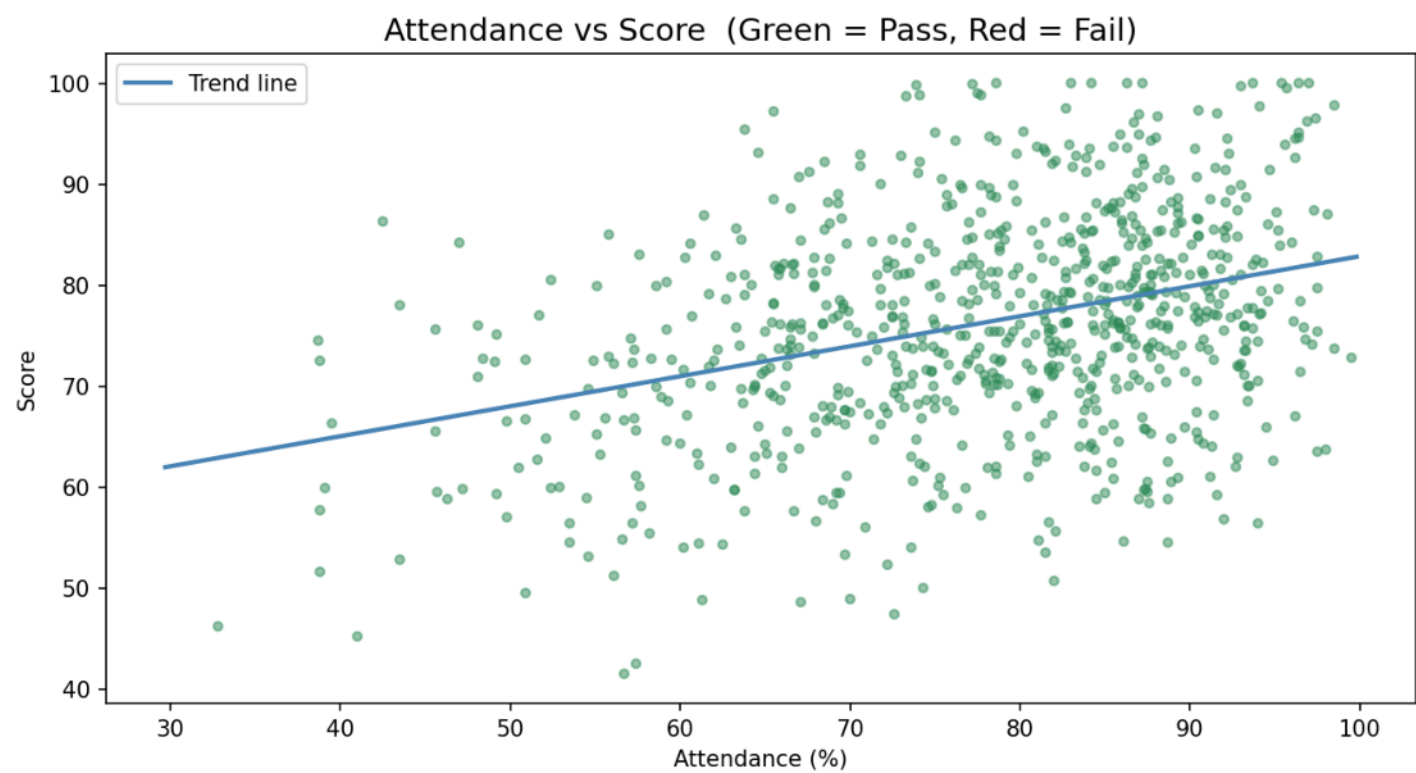


Figure 5

Study Hours vs Score

Study hours per day show a moderate positive correlation with scores ($r = 0.270$). Students who study more tend to perform better, though the link is not as strong as attendance. Scatter around the trend line indicates that study quality and teaching effectiveness also play important roles.

Study Hrs-Score r	Avg Study Hours	Recommended Min.
0.270	4.0 hrs / day	3 hrs / day

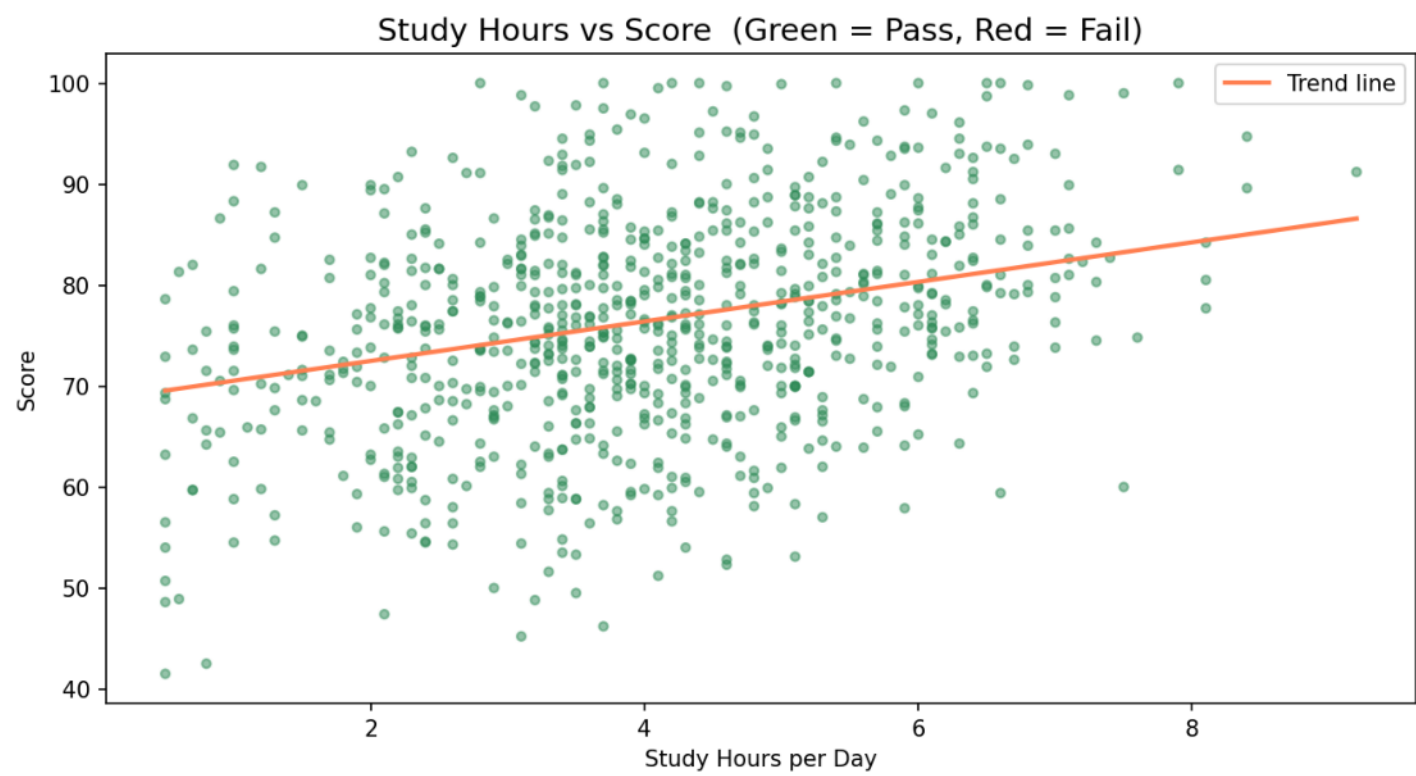


Figure 6

Average Score by Gender across Subjects

Grouped bars compare average scores for female and male students within each subject. Differences are generally small, suggesting performance is not strongly influenced by gender in this dataset. Any observable gap may warrant further investigation into teaching dynamics.

Genders	Subjects
Female / Male	5

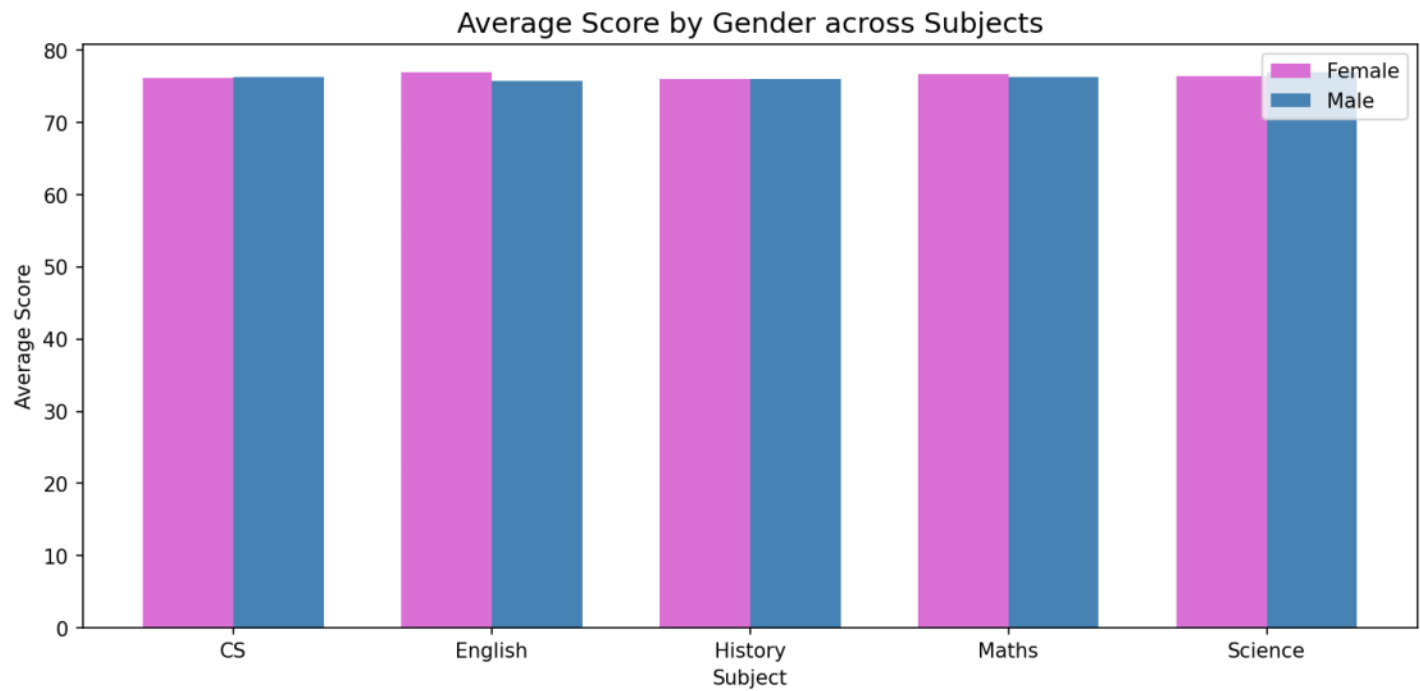


Figure 7

Pass Rate by Section

Section-level pass rates reveal disparities that subject averages can mask. Section C has the lowest pass rate, suggesting students in this group face specific challenges. High-performing sections serve as benchmarks for practical improvements.

Highest	Lowest
B (100.0%)	Section C (99.8%)

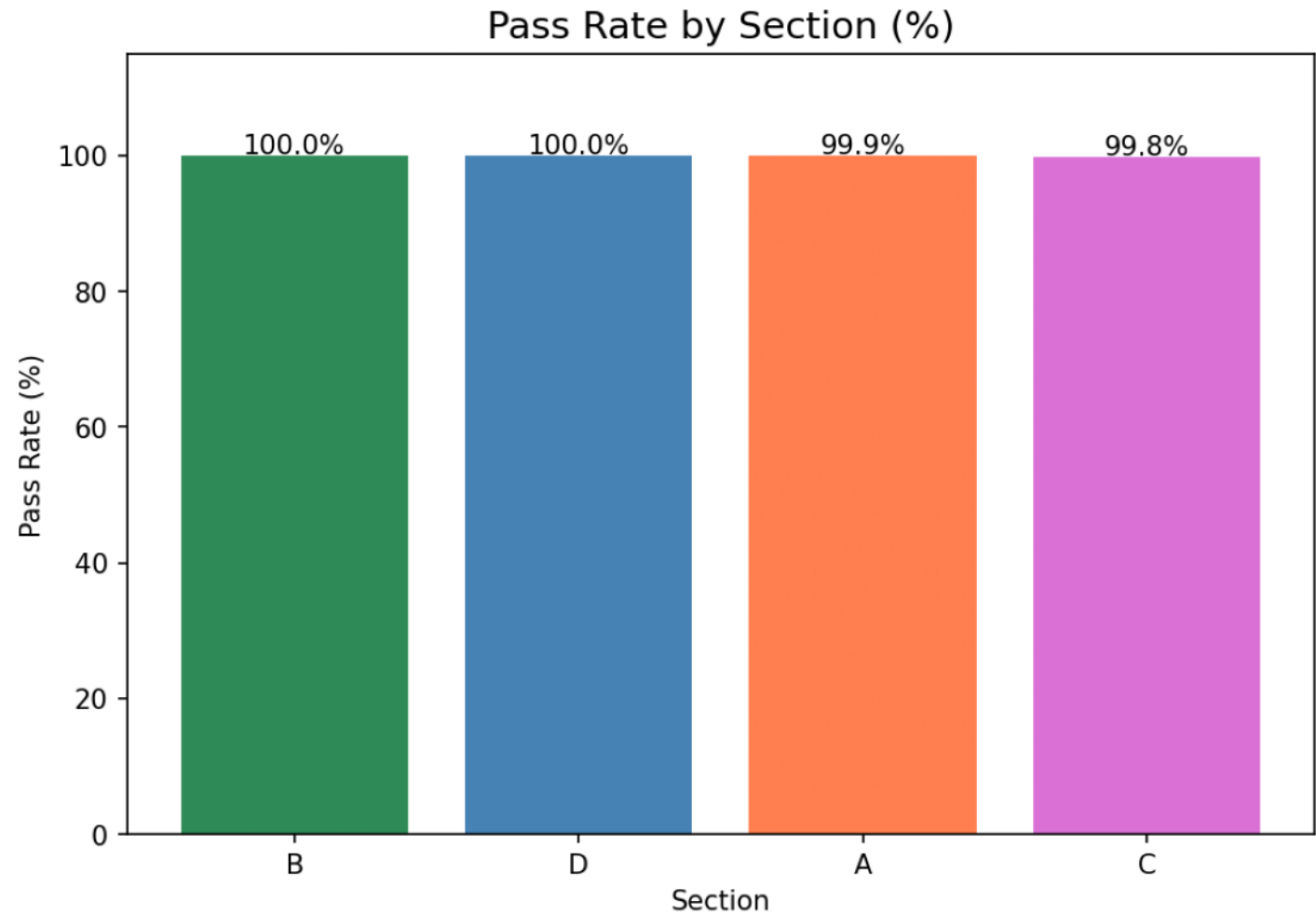


Figure 8

Correlation Heatmap

Attendance has the strongest link with score ($r = 0.338$), followed by study hours ($r = 0.270$). Attendance and study hours themselves show near-zero correlation — they are independent factors that each need to be addressed separately in any improvement strategy.

Attendance-Score r	Study Hrs-Score r	Attend.-Study r
0.338	0.270	-0.005

